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-- Q1

```
CREATE TABLE MEMBERSHIP (  
    MEM_NUM      INT          PRIMARY KEY,  
    MEM_FNAME    VARCHAR(30),  
    MEM_LNAME    VARCHAR(30),  
    MEM_STREET   VARCHAR(60),  
    MEM_CITY     VARCHAR(30),  
    MEM_STATE    CHAR(2),  
    MEM_ZIP      CHAR(10),  
    MEM_BALANCE  DECIMAL(8,2)  
);  
  
CREATE TABLE PRICE (  
    PRICE_CODE    INT          PRIMARY KEY,  
    PRICE_DESCRIPTION VARCHAR(30),  
    PRICE_RENTFEE  DECIMAL(6,2),  
    PRICE_DAILYLATEFEE DECIMAL(6,2)  
);  
  
CREATE TABLE MOVIE (  
    MOVIE_NUM     INT          PRIMARY KEY,  
    MOVIE_TITLE   VARCHAR(100),  
    MOVIE_YEAR    INT,  
    MOVIE_COST    DECIMAL(8,2),  
    MOVIE_GENRE   VARCHAR(20),  
    PRICE_CODE    INT,  
    CONSTRAINT fk_movie_price FOREIGN KEY (PRICE_CODE) REFERENCES PRICE(PRICE_CODE)  
);  
  
CREATE TABLE VIDEO (  
    VID_NUM      INT PRIMARY KEY,  
    VID_INDATE   DATE,  
    MOVIE_NUM     INT NOT NULL,  
    CONSTRAINT fk_video_movie FOREIGN KEY (MOVIE_NUM) REFERENCES MOVIE(MOVIE_NUM)  
);  
  
CREATE TABLE RENTAL (  
    RENT_NUM     INT PRIMARY KEY,  
    RENT_DATE    DATE,  
    MEM_NUM      INT,  
    CONSTRAINT fk_rental_member FOREIGN KEY (MEM_NUM) REFERENCES MEMBERSHIP(MEM_NUM)  
);  
  
CREATE TABLE DETAILRENTAL (  
    RENT_NUM      INT,  
    VID_NUM       INT,  
    DETAIL_FEE    DECIMAL(6,2),  
    DETAIL_DUEDATE DATE,
```

```

DETAIL_RETURNDATE    DATE,
DETAIL_DAILYLATEFEE  DECIMAL(6,2),
PRIMARY KEY (RENT_NUM, VID_NUM),
CONSTRAINT fk_dr_rental FOREIGN KEY (RENT_NUM) REFERENCES RENTAL(RENT_NUM),
CONSTRAINT fk_dr_video  FOREIGN KEY (VID_NUM)  REFERENCES VIDEO(VID_NUM)
);

```

RESULT:

Tables created: MEMBERSHIP, PRICE, MOVIE, VIDEO, RENTAL, DETAILRENTAL

-- Q2

```

INSERT INTO MEMBERSHIP VALUES
(102,'Tami','Dawson','2632 Takli Circle','Norene','TN','37136',11),
(103,'Curt','Knight','4025 Cornell Court','Flatgap','KY','41219',6),
(104,'Jamal','Melendez','788 East 145th Avenue','Quebeck','TN','38579',0),
(105,'Iva','Mcclain','6045 Musket Ball Circle','Summit','KY','42783',15),
(106,'Miranda','Parks','4469 Maxwell Place','Germantown','TN','38183',0),
(107,'Rosario','Elliott','7578 Danner Avenue','Columbia','TN','38402',5),
(108,'Mattie','Guy','4390 Evergreen Street','Lily','KY','40740',0),
(109,'Clint','Ochoa','1711 Elm Street','Greeneville','TN','37745',10),
(110,'Lewis','Rosales','4524 Southwind Circle','Counce','TN','38326',0),
(111,'Stacy','Mann','2789 East Cook Avenue','Murfreesboro','TN','37132',8),
(112,'Luis','Trujillo','7267 Melvin Avenue','Heiskell','TN','37754',3),
(113,'Minnie','Gonzales','6430 Vasili Drive','Williston','TN','38076',0);

```

```

INSERT INTO PRICE VALUES
(1,'Standard',2.00,1.00),
(2,'New Release',3.50,3.00),
(3,'Discount',1.50,1.00),
(4,'Weekly Special',1.00,0.50);

```

```

INSERT INTO MOVIE VALUES
(1234,'The Cesar Family Christmas',2014,39.95,'FAMILY',2),
(1235,'Smokey Mountain Wildlife',2011,59.95,'ACTION',1),
(1236,'Richard Goodhope',2015,59.95,'DRAMA',2),
(1237,'Beatnik Fever',2014,29.95,'COMEDY',2),
(1238,'Constant Companion',2015,89.95,'DRAMA',NULL),
(1239,'Where Hope Dies',2005,25.49,'DRAMA',3),
(1245,'Time to Burn',2012,45.49,'ACTION',1),
(1246,'What He Doesn\'t Know',2013,58.29,'COMEDY',1);

```

```

INSERT INTO VIDEO VALUES
(54321,'2015-06-18',1234),
(54324,'2015-06-18',1234),
(54325,'2015-06-18',1234),
(34341,'2014-01-22',1235),
(34342,'2014-01-22',1235),
(34366,'2016-03-02',1236),
(34367,'2016-03-02',1236),
(34368,'2016-03-02',1236),
(34369,'2016-03-02',1236),
(44392,'2015-10-21',1237),
(44397,'2015-10-21',1237),

```

```
(59237,'2016-02-14',1237),
(61388,'2014-01-25',1239),
(61353,'2013-01-28',1245),
(61354,'2013-01-28',1245),
(61367,'2015-07-30',1246),
(61369,'2015-07-30',1246);
```

INSERT INTO RENTAL VALUES

```
(1001,'2016-03-01',103),
(1002,'2016-03-01',105),
(1003,'2016-03-02',102),
(1004,'2016-03-02',110),
(1005,'2016-03-02',111),
(1006,'2016-03-02',107),
(1007,'2016-03-02',104),
(1008,'2016-03-03',105),
(1009,'2016-03-03',111);
```

INSERT INTO DETAILRENTAL VALUES

```
(1001,34342,2.00,'2016-03-04','2016-03-02',NULL),
(1001,61353,2.00,'2016-03-04','2016-03-03',1.00),
(1002,59237,3.50,'2016-03-04','2016-03-04',3.00),
(1003,54325,3.50,'2016-03-04','2016-03-09',3.00),
(1003,61369,2.00,'2016-03-06','2016-03-09',1.00),
(1003,61388,0.00,'2016-03-06','2016-03-09',1.00),
(1004,44392,3.50,'2016-03-05','2016-03-07',3.00),
(1004,34367,3.50,'2016-03-05','2016-03-07',3.00),
(1004,34341,2.00,'2016-03-07','2016-03-07',1.00),
(1005,34342,2.00,'2016-03-07','2016-03-05',1.00),
(1005,44397,3.50,'2016-03-05','2016-03-05',3.00),
(1006,34366,3.50,'2016-03-05','2016-03-04',3.00),
(1006,61367,2.00,'2016-03-07',NULL,1.00),
(1007,34368,3.50,'2016-03-05',NULL,3.00),
(1008,34369,3.50,'2016-03-05','2016-03-05',3.00),
(1009,54324,3.50,'2016-03-05',NULL,3.00),
(1001,34366,3.50,'2016-03-04','2016-03-02',3.00);
```

RESULT: All rows inserted successfully into all 6 tables.

-- Q3

-- BEFORE:

```
SELECT MOVIE_NUM, MOVIE_TITLE, MOVIE_YEAR FROM MOVIE WHERE MOVIE_NUM = 1245;
MOVIE_NUM  MOVIE_TITLE      MOVIE_YEAR
1245       Time to Burn      2012
```

UPDATE MOVIE SET MOVIE_YEAR = 2010 WHERE MOVIE_NUM = 1245;

-- AFTER:

```
SELECT MOVIE_NUM, MOVIE_TITLE, MOVIE_YEAR FROM MOVIE WHERE MOVIE_NUM = 1245;
MOVIE_NUM  MOVIE_TITLE      MOVIE_YEAR
1245       Time to Burn      2010
```

-- Q4

-- BEFORE:

```
SELECT MOVIE_NUM, MOVIE_TITLE, MOVIE_GENRE, PRICE_CODE FROM MOVIE WHERE MOVIE_GENRE = 'ACTION';
```

MOVIE_NUM	MOVIE_TITLE	MOVIE_GENRE	PRICE_CODE
1235	Smokey Mountain Wildlife	ACTION	1
1245	Time to Burn	ACTION	1

```
UPDATE MOVIE SET PRICE_CODE = 3 WHERE MOVIE_GENRE = 'ACTION';
```

-- AFTER:

```
SELECT MOVIE_NUM, MOVIE_TITLE, MOVIE_GENRE, PRICE_CODE FROM MOVIE WHERE MOVIE_GENRE = 'ACTION';
```

MOVIE_NUM	MOVIE_TITLE	MOVIE_GENRE	PRICE_CODE
1235	Smokey Mountain Wildlife	ACTION	3
1245	Time to Burn	ACTION	3

-- Q5

-- BEFORE:

```
SELECT * FROM PRICE;
```

PRICE_CODE	PRICE_DESCRIPTION	PRICE_RENTFEE	PRICE_DAILYLATEFEE
1	Standard	2.00	1.00
2	New Release	3.50	3.00
3	Discount	1.50	1.00
4	Weekly Special	1.00	0.50

```
UPDATE PRICE SET PRICE_RENTFEE = PRICE_RENTFEE + 0.50;
```

-- AFTER:

```
SELECT * FROM PRICE;
```

PRICE_CODE	PRICE_DESCRIPTION	PRICE_RENTFEE	PRICE_DAILYLATEFEE
1	Standard	2.50	1.00
2	New Release	4.00	3.00
3	Discount	2.00	1.00
4	Weekly Special	1.50	0.50

-- Q6

```
SELECT MOVIE_TITLE, MOVIE_YEAR, MOVIE_GENRE FROM MOVIE;
```

RESULT:

Movie_Title	Movie_Year	Movie_Genre
The Cesar Family Christmas	2014	FAMILY
Smokey Mountain Wildlife	2011	ACTION
Richard Goodhope	2015	DRAMA
Beatnik Fever	2014	COMEDY
Constant Companion	2015	DRAMA
Where Hope Dies	2005	DRAMA
Time to Burn	2010	ACTION
What He Doesn't Know	2013	COMEDY

-- Q7

```
SELECT MOVIE_YEAR, MOVIE_TITLE, MOVIE_COST FROM MOVIE ORDER BY MOVIE_YEAR DESC;
```

RESULT:

Movie_Year	Movie_Title	Movie_Cost
2015	Richard Goodhope	59.95
2015	Constant Companion	89.95
2014	The Cesar Family Christmas	39.95
2014	Beatnik Fever	29.95
2013	What He Doesn't Know	58.29
2011	Smokey Mountain Wildlife	59.95
2010	Time to Burn	45.49
2005	Where Hope Dies	25.49

-- Q8

```
SELECT MOVIE_TITLE, MOVIE_YEAR, MOVIE_GENRE  
FROM MOVIE  
ORDER BY MOVIE_GENRE ASC, MOVIE_YEAR DESC;
```

RESULT:

Movie_Title	Movie_Year	Movie_Genre
Smokey Mountain Wildlife	2011	ACTION
Time to Burn	2010	ACTION
Beatnik Fever	2014	COMEDY
What He Doesn't Know	2013	COMEDY
Richard Goodhope	2015	DRAMA
Constant Companion	2015	DRAMA
Where Hope Dies	2005	DRAMA
The Cesar Family Christmas	2014	FAMILY

-- Q9

```
SELECT MOVIE_NUM, MOVIE_TITLE, PRICE_CODE FROM MOVIE WHERE MOVIE_TITLE LIKE 'R%';
```

RESULT:

Movie_Num	Movie_Title	Price_Code
1236	Richard Goodhope	2

-- Q10

```
SELECT MOVIE_TITLE, MOVIE_YEAR, MOVIE_COST  
FROM MOVIE  
WHERE MOVIE_TITLE LIKE '%hope%'  
ORDER BY MOVIE_TITLE ASC;
```

RESULT:

Movie_Title	Movie_Year	Movie_Cost
Richard Goodhope	2015	59.95
Where Hope Dies	2005	25.49

-- Q11

```
SELECT MOVIE_TITLE, MOVIE_YEAR, MOVIE_GENRE FROM MOVIE WHERE MOVIE_GENRE = 'ACTION';
```

RESULT:

Movie_Title	Movie_Year	Movie_Genre
Smokey Mountain Wildlife	2011	ACTION
Time to Burn	2010	ACTION

-- Q12

```
SELECT MOVIE_NUM, MOVIE_TITLE, MOVIE_COST FROM MOVIE WHERE MOVIE_COST > 40;
```

RESULT:

Movie_Num	Movie_Title	Movie_Cost
1235	Smokey Mountain Wildlife	59.95
1236	Richard Goodhope	59.95
1238	Constant Companion	89.95
1245	Time to Burn	45.49
1246	What He Doesn't Know	58.29

-- Q13

```
SELECT MOVIE_NUM, MOVIE_TITLE, MOVIE_COST, MOVIE_GENRE
FROM MOVIE
WHERE (MOVIE_GENRE = 'ACTION' OR MOVIE_GENRE = 'COMEDY') AND MOVIE_COST < 50
ORDER BY MOVIE_GENRE ASC;
```

RESULT:

Movie_Num	Movie_Title	Movie_Cost	Movie_Genre
1245	Time to Burn	45.49	ACTION
1237	Beatnik Fever	29.95	COMEDY

-- Q14

```
SELECT MEM_NUM, MEM_FNAME, MEM_LNAME, MEM_STREET, MEM_STATE, MEM_BALANCE
FROM MEMBERSHIP
WHERE MEM_STATE = 'TN' AND MEM_BALANCE < 5 AND MEM_STREET LIKE '%Avenue';
```

RESULT:

Mem_Num	Mem_FName	Mem_LName	Mem_Street	Mem_State	Mem_Balance
104	Jamal	Melendez	788 East 145th Avenue	TN	0.00
112	Luis	Trujillo	7267 Melvin Avenue	TN	3.00

-- Q15

```
SELECT MOVIE_GENRE, COUNT(*) AS 'Number of Movies'
FROM MOVIE
GROUP BY MOVIE_GENRE;
```

RESULT:

Movie_Genre	Number of Movies
ACTION	2
COMEDY	2
DRAMA	3
FAMILY	1

-- Q16

SELECT AVG(MOVIE_COST) AS 'Average Movie Cost' FROM MOVIE;

RESULT:

Average Movie Cost
51.1275

-- Q17

SELECT MOVIE_GENRE, AVG(MOVIE_COST) AS 'Average Cost'
FROM MOVIE
GROUP BY MOVIE_GENRE;

RESULT:

Movie_Genre	Average Cost
ACTION	52.72
COMEDY	44.12
DRAMA	58.46
FAMILY	39.95

-- Q18

SELECT m.MOVIE_TITLE, m.MOVIE_GENRE, p.PRICE_DESCRIPTION, p.PRICE_RENTFEE
FROM MOVIE m
JOIN PRICE p ON m.PRICE_CODE = p.PRICE_CODE;

RESULT:

Movie_Title	Movie_Genre	Price_Description	Price_RentFee
The Cesar Family Christmas	FAMILY	New Release	4.00
Smokey Mountain Wildlife	ACTION	Discount	2.00
Richard Goodhope	DRAMA	New Release	4.00
Beatnik Fever	COMEDY	New Release	4.00
Where Hope Dies	DRAMA	Discount	2.00
Time to Burn	ACTION	Discount	2.00
What He Doesn't Know	COMEDY	Standard	2.50

-- Q19

SELECT m.MOVIE_GENRE, AVG(p.PRICE_RENTFEE) AS 'Average Rental Fee'
FROM MOVIE m
JOIN PRICE p ON m.PRICE_CODE = p.PRICE_CODE
GROUP BY m.MOVIE_GENRE;

RESULT:

Movie_Genre	Average Rental Fee
ACTION	2.00
COMEDY	3.25
DRAMA	3.00
FAMILY	4.00

-- Q20

```
SELECT m.MOVIE_TITLE, ROUND(m.MOVIE_COST / p.PRICE_RENTFEE, 2) AS 'Breakeven Rentals'
FROM MOVIE m
JOIN PRICE p ON m.PRICE_CODE = p.PRICE_CODE;
```

RESULT:

Movie_Title	Breakeven Rentals
The Cesar Family Christmas	9.99
Smokey Mountain Wildlife	29.98
Richard Goodhope	14.99
Beatnik Fever	7.49
Where Hope Dies	12.75
Time to Burn	22.75
What He Doesn't Know	23.32

-- Q21

```
SELECT MOVIE_TITLE, MOVIE_YEAR FROM MOVIE WHERE PRICE_CODE IS NOT NULL;
```

RESULT:

Movie_Title	Movie_Year
The Cesar Family Christmas	2014
Smokey Mountain Wildlife	2011
Richard Goodhope	2015
Beatnik Fever	2014
Where Hope Dies	2005
Time to Burn	2010
What He Doesn't Know	2013

-- Q22

```
SELECT MOVIE_TITLE, MOVIE_GENRE, MOVIE_COST
FROM MOVIE
WHERE MOVIE_COST BETWEEN 44.99 AND 49.99;
```

RESULT:

Movie_Title	Movie_Genre	Movie_Cost
Time to Burn	ACTION	45.49

-- Q23

```
SELECT m.MOVIE_TITLE, p.PRICE_DESCRIPTION, p.PRICE_RENTFEE, m.MOVIE_GENRE
```



```

FROM MOVIE m
JOIN PRICE p ON m.PRICE_CODE = p.PRICE_CODE
WHERE m.MOVIE_GENRE IN ('FAMILY', 'COMEDY', 'DRAMA');

```

RESULT:

Movie_Title	Price_Description	Price_RentFee	Movie_Genre
The Cesar Family Christmas	New Release	4.00	FAMILY
Richard Goodhope	New Release	4.00	DRAMA
Beatnik Fever	New Release	4.00	COMEDY
Where Hope Dies	Discount	2.00	DRAMA
What He Doesn't Know	Standard	2.50	COMEDY

-- Q24

```

SELECT DISTINCT mb.MEM_NUM, mb.MEM_FNAME, mb.MEM_LNAME, mb.MEM_BALANCE
FROM MEMBERSHIP mb
JOIN RENTAL r ON mb.MEM_NUM = r.MEM_NUM;

```

RESULT:

Mem_Num	Mem_FName	Mem_LName	Mem_Balance
102	Tami	Dawson	11.00
103	Curt	Knight	6.00
104	Jamal	Melendez	0.00
105	Iva	Mcclain	15.00
107	Rosario	Elliott	5.00
110	Lewis	Rosales	0.00
111	Stacy	Mann	8.00

-- Q25

```

SELECT MIN(mb.MEM_BALANCE) AS 'Minimum Balance',
       MAX(mb.MEM_BALANCE) AS 'Maximum Balance',
       AVG(mb.MEM_BALANCE) AS 'Average Balance'
FROM MEMBERSHIP mb
JOIN RENTAL r ON mb.MEM_NUM = r.MEM_NUM;

```

RESULT:

Minimum Balance	Maximum Balance	Average Balance
0.00	15.00	6.43

-- Q26

```

SELECT dr.RENT_NUM, r.RENT_DATE, dr.VID_NUM, m.MOVIE_TITLE,
       dr.DETAIL_DUEDATE, dr.DETAIL_RETURNDATE
FROM DETAILRENTAL dr
JOIN RENTAL r ON dr.RENT_NUM = r.RENT_NUM
JOIN VIDEO v ON dr.VID_NUM = v.VID_NUM
JOIN MOVIE m ON v.MOVIE_NUM = m.MOVIE_NUM
WHERE dr.DETAIL_RETURNDATE > dr.DETAIL_DUEDATE
ORDER BY dr.RENT_NUM, m.MOVIE_TITLE;

```

RESULT:

Rent_Num	Rent_Date	Vid_Num	Movie_Title	Detail_DueDate	Detail_ReturnDate
1003	2016-03-02	54325	The Cesar Family Christmas	2016-03-04	2016-03-09
1003	2016-03-02	61369	What He Doesn't Know	2016-03-06	2016-03-09
1003	2016-03-02	61388	Where Hope Dies	2016-03-06	2016-03-09
1004	2016-03-02	44392	Beatnik Fever	2016-03-05	2016-03-07
1004	2016-03-02	34367	Richard Goodhope	2016-03-05	2016-03-07

-- Q27

```

SELECT dr.RENT_NUM, r.RENT_DATE, m.MOVIE_TITLE, dr.DETAIL_FEE
FROM DETAILRENTAL dr
JOIN RENTAL r ON dr.RENT_NUM = r.RENT_NUM
JOIN VIDEO v ON dr.VID_NUM = v.VID_NUM
JOIN MOVIE m ON v.MOVIE_NUM = m.MOVIE_NUM
WHERE dr.DETAIL_RETURNDATE <= dr.DETAIL_DUEDATE;

```

RESULT:

Rent_Num	Rent_Date	Movie_Title	Detail_Fee
1001	2016-03-01	Smokey Mountain Wildlife	2.00
1001	2016-03-01	Richard Goodhope	3.50
1001	2016-03-01	Time to Burn	2.00
1002	2016-03-01	Beatnik Fever	3.50
1004	2016-03-02	Smokey Mountain Wildlife	2.00
1005	2016-03-02	Smokey Mountain Wildlife	2.00
1005	2016-03-02	Beatnik Fever	3.50
1006	2016-03-02	Richard Goodhope	3.50
1008	2016-03-03	Richard Goodhope	3.50

-- Q28

```

SELECT m.MOVIE_NUM,
       m.MOVIE_GENRE,
       ROUND(AVG(m2.MOVIE_COST), 2) AS 'Average Cost',
       m.MOVIE_COST AS 'Movie Cost',
       ROUND((m.MOVIE_COST - AVG(m2.MOVIE_COST)) / AVG(m2.MOVIE_COST) * 100, 2) AS 'Percent
Difference'
FROM MOVIE m
JOIN MOVIE m2 ON m.MOVIE_GENRE = m2.MOVIE_GENRE
GROUP BY m.MOVIE_NUM, m.MOVIE_GENRE, m.MOVIE_COST
ORDER BY m.MOVIE_NUM;

```

RESULT:

Movie_Num	Movie_Genre	Average Cost	Movie_Cost	Percent Difference
1234	FAMILY	39.95	39.95	0.00
1235	ACTION	52.72	59.95	13.71
1236	DRAMA	58.46	59.95	2.54
1237	COMEDY	44.12	29.95	-32.12
1238	DRAMA	58.46	89.95	53.86
1239	DRAMA	58.46	25.49	-56.40
1245	ACTION	52.72	45.49	-13.71
1246	COMEDY	44.12	58.29	32.12